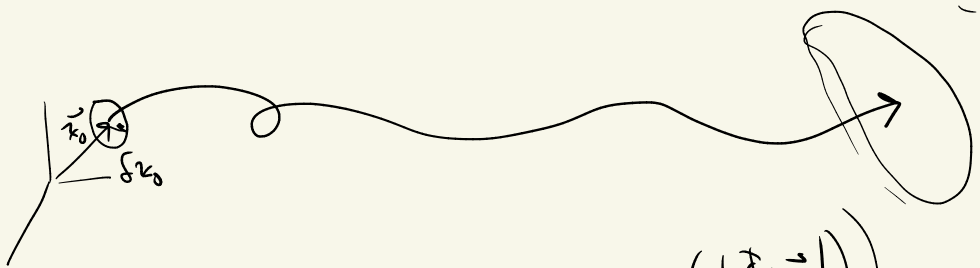
Non-Periodic Trajectories:

How can we characterize them? They extend to $t \to \pm\infty$

- Lyapunov Characteristic Exponents do this characterization for us.
- Generalize the idea of stability to arbitrary trajectories.

Def: The LCE of a trajectory $\vec{x}(t) = \varphi(t; \vec{x}_0, t_0)$

with the attendant S.T.M. $\delta\vec{x}(t) = \Phi(t, t_0; \vec{x}_0) \cdot \delta\vec{x}_0$



$$\chi = \max_{\vec{u}}\left(\lim_{T\to\infty}\frac{1}{T}\ln\left(\frac{|\Phi\cdot\vec{u}|}{|\vec{u}|}\right)\right) \quad ; \quad |\vec{u}| = \sqrt{\vec{u}\cdot\vec{u}}$$

IF $\chi > 0$, the trajectory is unstable (n'bring traj. exponentially diverge)

IF $\chi = 0$, " " " stable (" " "hang out" near each other)

For a fixed T, consider

$$\max_{\vec{u}} \ln\left[\frac{|\Phi\cdot\vec{u}|}{|\vec{u}|}\right] \sim \max_{\vec{u}} \frac{\sqrt{\vec{u}^T\cdot\Phi^T\Phi\cdot\vec{u}}}{\sqrt{\vec{u}^T\vec{u}}} \sim \max_{|\vec{u}|=1} \vec{u}\cdot\Phi^T\Phi\cdot\vec{u}$$

Constrained cost function:

$$\boxed{J = \vec{u}^T\cdot\Phi^T\Phi\cdot\vec{u} - \mu(\vec{u}\cdot\vec{u}-1)}$$

$$\frac{\partial J}{\partial \vec{u}} = 2\left[\Phi^T\Phi\cdot\vec{u} - \mu\vec{u}\right] = \boxed{2\underbrace{\left[\Phi^T\Phi - \mu I\right]}\cdot\vec{u} = 0}$$

$$\frac{\partial J}{\partial \mu} = \vec{u}\cdot\vec{u} - 1 = 0 \Rightarrow |\vec{u}| = 1$$

$$\left[\Phi^T\Phi - \mu I\right]\cdot\vec{a} = \vec{0} \quad ; \quad \text{Find ev's + } \vec{ev}\text{'s of } \underbrace{\Phi^T\Phi}$$

"Cauchy Green Strain Tensor"

$$\boxed{\delta x^2 = \delta x_0^T \Phi^T \Phi \, \delta x_0}$$

ev's $\Phi^T\Phi$? Related to ev's Φ, λ_i, $i=1,\dots,6$

ev's $\Phi^T\Phi$, $\boxed{\mu_i = \lambda_i \bar{\lambda}_i}$: IF $\lambda_i \in \mathbb{R}$, $\mu_i = \lambda_i^2$

$\lambda_i = e^{+i\theta}$, $\mu_i = \lambda_i\bar{\lambda}_i = 1$

$$\boxed{\chi = \lim_{T\to\infty} \frac{1}{T}\left(\max_{i} \ln\left(\lambda_i \bar{\lambda}_i\right)\right) \; ; \; \lambda_i \in ev(\Phi)}$$

Special Cases:

For an Eq. pt $\quad \lambda_i = e^{\alpha_i T \pm i\beta_i T}$; $\sqrt{\lambda_i \bar{\lambda}_i}^{\,T} = e^{\alpha_i T}$

$$\chi = \frac{1}{T} \max_i \ln\left[e^{\alpha_i T}\right] = \frac{1}{\cancel{T}} \alpha_{max} \cancel{T} = \underline{\alpha_{max}}$$

Eq pt $\quad \lambda_i = e^{\pm i\beta_i T}$; $\lambda_i \bar{\lambda}_i = 1$, $\ln(1) = 0$

$$\chi = 0.$$

Periodic orbits, $T = n T_{period} \to \infty$

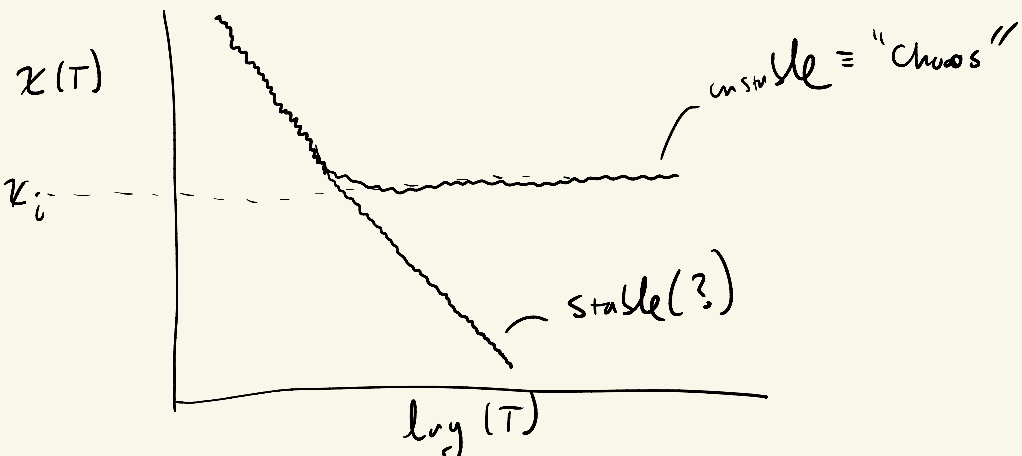
$$\chi = \max_{\hat{i}} \alpha_{\hat{i}}$$

Practical Approaches to Computation: Introduce nearby variation $\delta\vec{x}_0$

Can show that

$$\chi(T) \cong \frac{\ln\left|\varphi(T_j;\vec{x}_0+\delta\vec{x}_0,t_0)-\varphi(T_j;\vec{x}_0,t_0)\right|}{T\,|\delta\vec{x}_0|}_j \,.$$

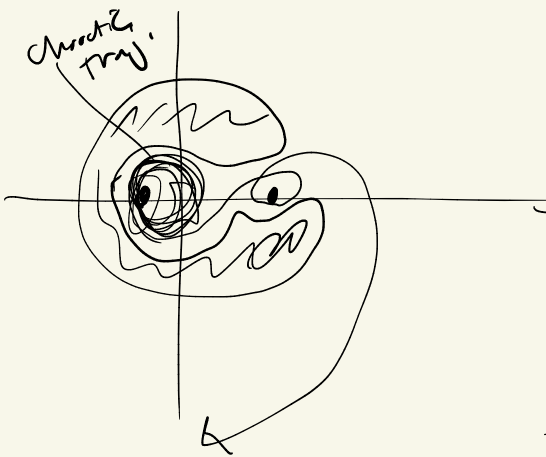
If plotted as a function of T we see trends that allow us to infer the LCE. Must choose $|\delta x_0| \ll 1$.



Not definitive

Finite Time LCEs, can describe local motion for a short period of time, but may still be "stable" as $T \to \pm\infty$.

CR3BP



There are many robust ways to compute LCEs, let me know if you need references.

Global Bounds on Trajectories